We would have understood one model chemical system fairly well, at least with respect to certain qualitative issues.

There are, however, thousands of distinct reaction networks that might, on one occasion or another, command our attention. Each has its own system of differential equations, perhaps more complicated by far than the system we have been considering. How, then, are we to proceed? It is clear that we cannot rely forever on purely ad hoc studies of whatever systems might present themselves for examination. Even if we cast aside the long-term enormity of such an undertaking, there are still two problems that must be faced in the short run. First, questions of the kind we have posed will, for the most part, be confronted by engineers and chemists, not mathematicians. Second, it is by no means clear that mathematicians, even the most expert, are currently in a position to provide much help. The fact is that even moderately large systems of nonlinear differential equations — in particular polynomial systems like those displayed in (1.7) — remain poorly understood* in general.

It seems to me that what is required is a rather broad-based theory of those systems of differential equations that derive from reaction networks, a theory which would in some sense cuts across the fine details of individual problems to provide qualitative information about large classes of systems all at once. Moreover, we would like the results of such a theory to be of the kind that engineers and chemists can use easily in addressing questions like those we have posed.

This seems like a lot to ask, and I should try to explain why a theory of the kind I have in mind should even be possible in principle. Although we shall also be interested in the more general situation, let me temporarily restrict my attention to reaction networks endowed with mass action kinetics. Thinking back to the source of the system (1.7), we recall that it derived in a rather orderly way from the network (1.6). In fact, we knew how to write down the appropriate

*Consider, for example, the remarkable complexity of the seemingly innocuous Lorenz system [L1], which is composed of three polynomial ordinary differential equations in three variables.